

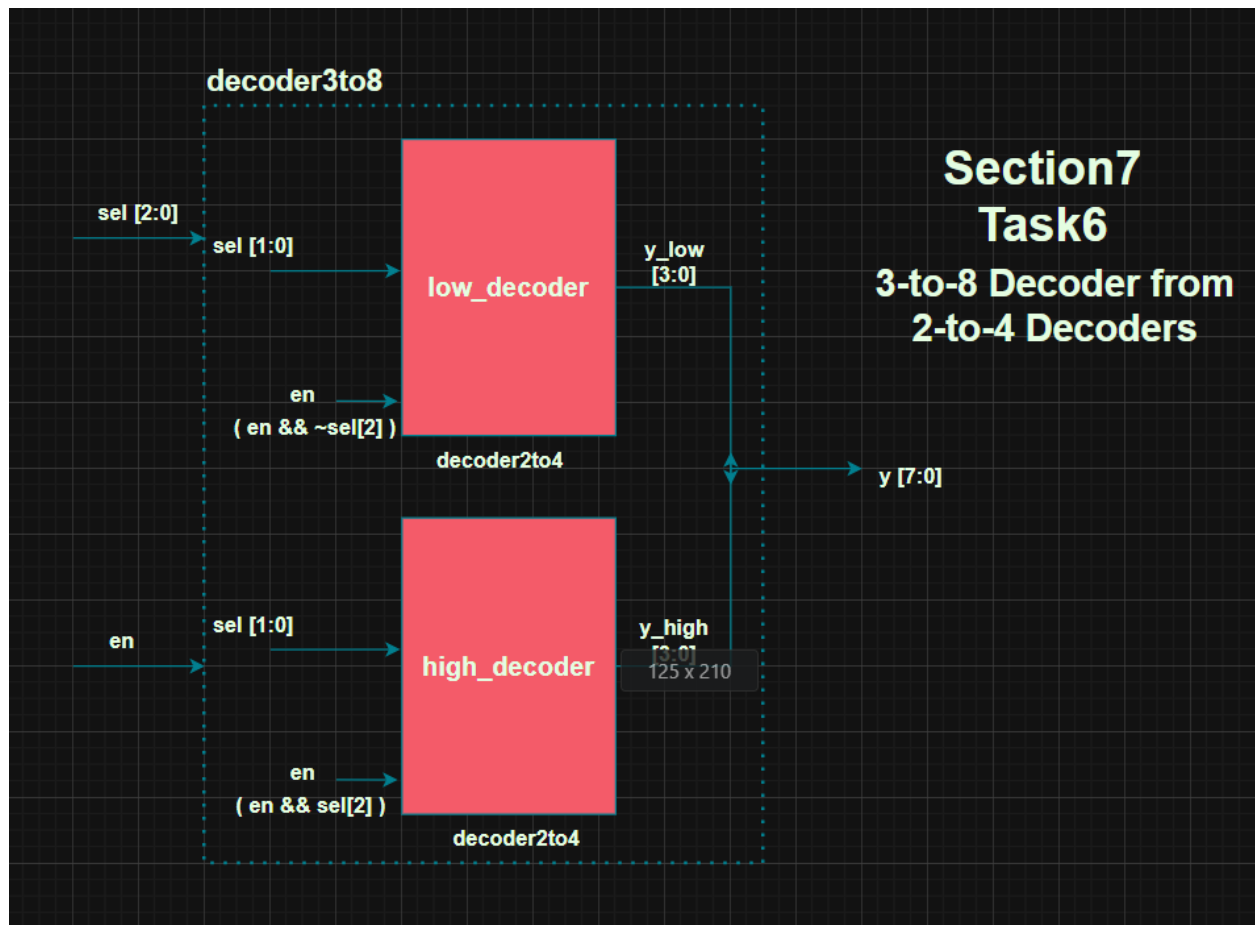
NAME: MUHAMMAD USMAN

REG. NO# 2024-EE-171

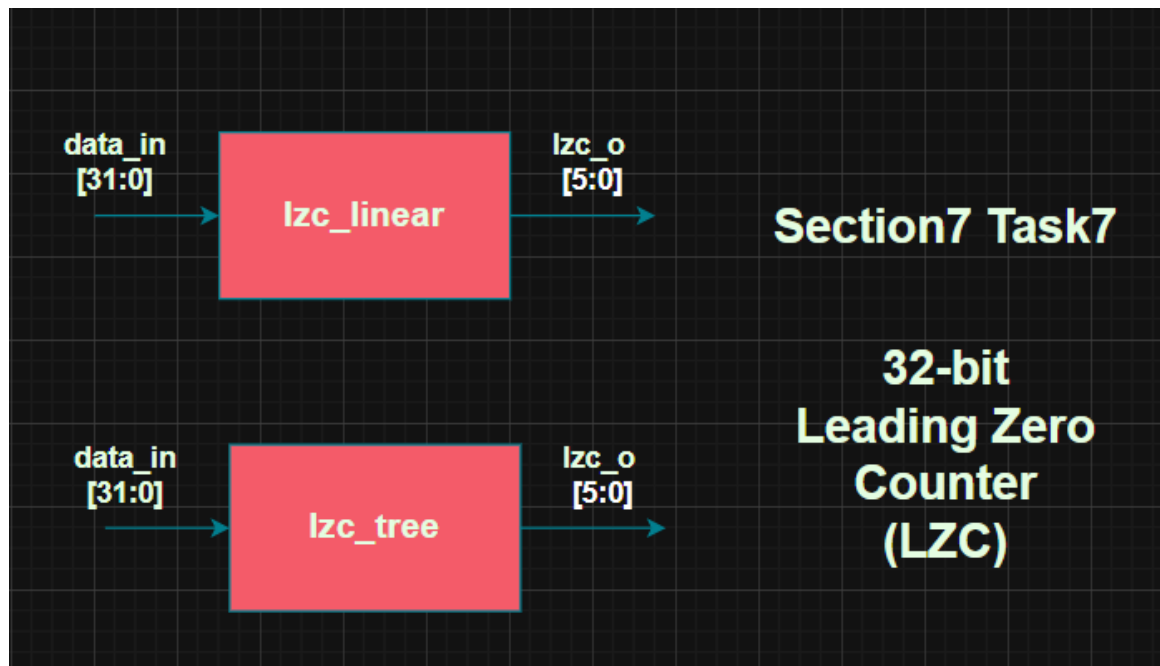
1-EDA Playground link of Task6: <https://edaplayground.com/x/aiKQ>

EDA Playground link of Task7: <https://edaplayground.com/x/Scdc>

2-Schematic Diagram:



Task6_schematic_diagram.png



Task7_schematic_diagram.png

3-Output Screenshot:

```

Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full64; Runtime version X-2025.06-SP1_Full64;  Jul 20 03:46 2026
[PASS] en=1 sel=0 (0) : y=1
[PASS] en=1 sel=1 (1) : y=10
[PASS] en=1 sel=10 (2) : y=100
[PASS] en=1 sel=11 (3) : y=1000
[PASS] en=0 sel=0 (0) : y=0
[PASS] en=0 sel=1 (1) : y=0
[PASS] en=0 sel=10 (2) : y=0
[PASS] en=0 sel=11 (3) : y=0
$finish called from file "testbench.sv", line 40.
$finish at simulation time          90
      V C S   S i m u l a t i o n   R e p o r t
Time: 90 ns
CPU Time:      0.490 seconds;      Data structure size:  0.0Mb
Mon Jul 20 03:46:47 2026
Finding VCD file...
./dump.vcd
[2026-07-20 07:46:48 UTC] Opening EPWave...
Done

```

Task6_decoder2to4_log_result.png

```

Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 20 03:45 2026
[PASS] en=1 sel=0 (0) : y=1
[PASS] en=1 sel=1 (1) : y=10
[PASS] en=1 sel=10 (2) : y=100
[PASS] en=1 sel=11 (3) : y=1000
[PASS] en=1 sel=100 (4) : y=10000
[PASS] en=1 sel=101 (5) : y=100000
[PASS] en=1 sel=110 (6) : y=1000000
[PASS] en=1 sel=111 (7) : y=10000000
[PASS] en=0 sel=0 (0) : y=0
[PASS] en=0 sel=1 (1) : y=0
[PASS] en=0 sel=10 (2) : y=0
[PASS] en=0 sel=11 (3) : y=0
[PASS] en=0 sel=100 (4) : y=0
[PASS] en=0 sel=101 (5) : y=0
[PASS] en=0 sel=110 (6) : y=0
[PASS] en=0 sel=111 (7) : y=0
$finish called from file "testbench.sv", line 40.
$finish at simulation time 170
V C S S i m u l a t i o n R e p o r t
Time: 170 ns
CPU Time: 0.560 seconds; Data structure size: 0.0Mb
Mon Jul 20 03:45:23 2026
Finding VCD file...
./dump.vcd
[2026-07-20 07:45:23 UTC] Opening EPWave...
Done

```

Task6_log_result.png

```

Log Share
Chronologic VCS simulator copyright 1991-2025
Contains Synopsys proprietary information.
Compiler version X-2025.06-SP1_Full164; Runtime version X-2025.06-SP1_Full164; Jul 20 06:18 2026
[PASS] in=0 zeros=32
[PASS] in=1 zeros=31
[PASS] in=8000 zeros=16
[PASS] in=80000000 zeros=0
[PASS] in=5e25336 zeros=5
[PASS] in=57dea13c zeros=1
[PASS] in=be7fe77d zeros=0
[PASS] in=30aa33e2 zeros=2
[PASS] in=279440b zeros=6
[PASS] in=b9b381df zeros=0
[PASS] in=f350a940 zeros=0
[PASS] in=d51778f7 zeros=0
[PASS] in=6d7e79a6 zeros=1
[PASS] in=30d4b41b zeros=2
[PASS] in=25b9db5 zeros=6
[PASS] in=82e394fa zeros=0
[PASS] in=19bc7a4e zeros=3
[PASS] in=5a98a215 zeros=1
[PASS] in=7985c17d zeros=1
[PASS] in=e43a5f72 zeros=0
[PASS] in=ffb4196 zeros=0
[PASS] in=217cb731 zeros=2
[PASS] in=9df554c4 zeros=0
[PASS] in=cac8ccaa zeros=0
[PASS] in=f58bc0c4 zeros=0
[PASS] in=11ab26cf zeros=3
[PASS] in=38646d4f zeros=2
[PASS] in=e8478af4 zeros=0
[PASS] in=7337dd17 zeros=1
[PASS] in=76264a88 zeros=1
[PASS] in=2f9fc9f1 zeros=2
[PASS] in=2da1e52c zeros=2
[PASS] in=94d34dce zeros=0
[PASS] in=9bee7505 zeros=0
[PASS] in=93fee0cb zeros=0
[PASS] in=3af0db8c zeros=2
[PASS] in=ac13f71a zeros=0
[PASS] in=8ef14f37 zeros=0
[PASS] in=4c23c660 zeros=1

```

Task7_log_result1.png

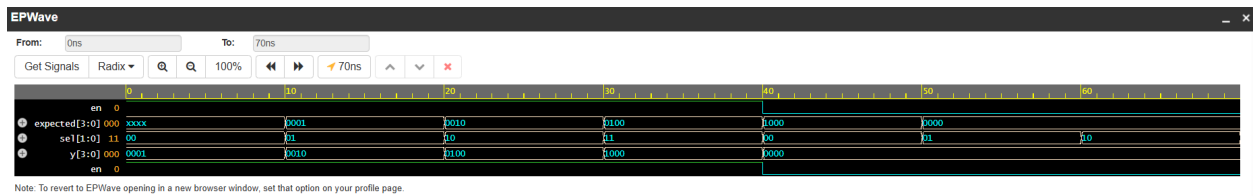
```

Log Share
[PASS] in=9bee7505 zeros=0
[PASS] in=93fee0cb zeros=0
[PASS] in=3af0db8c zeros=2
[PASS] in=ac13f71a zeros=0
[PASS] in=8ef14f37 zeros=0
[PASS] in=4c23c660 zeros=1
[PASS] in=1ac5a95f zeros=3
[PASS] in=bb91ab7a zeros=0
[PASS] in=b1487ba2 zeros=0
[PASS] in=8a4b2cf0 zeros=0
[PASS] in=f7fb42c9 zeros=0
[PASS] in=7dc155dc zeros=1
[PASS] in=4a089e41 zeros=1
[PASS] in=84285d3f zeros=0
[PASS] in=2890cc12 zeros=2
[PASS] in=cded48f4 zeros=0
[PASS] in=a8be83df zeros=0
[PASS] in=53a24ec5 zeros=1
[PASS] in=c8f8028d7 zeros=0
[PASS] in=92d2a694 zeros=0
[PASS] in=a5568088 zeros=0
[PASS] in=c4f6001 zeros=4
[PASS] in=867b8d31 zeros=0
[PASS] in=26b88d29 zeros=2
[PASS] in=65a127d6 zeros=1
[PASS] in=14ecc7f4 zeros=3
[PASS] in=c20e5ad9 zeros=0
[PASS] in=3d77c14f zeros=2
[PASS] in=fc69c200 zeros=0
[PASS] in=fe1f66dd zeros=0
[PASS] in=13f29dd2 zeros=3
All tests passed!
$finish called from file "testbench.sv", line 55.
$finish at simulation time 64000
V C S S i m u l a t i o n R e p o r t
Time: 64000 ps
CPU Time: 0.570 seconds; Data structure size: 0.0Mb
Mon Jul 20 06:18:46 2026
Finding VCD file...
./dump.vcd
[2026-07-20 10:18:47 UTC] Opening EPWave...
Done

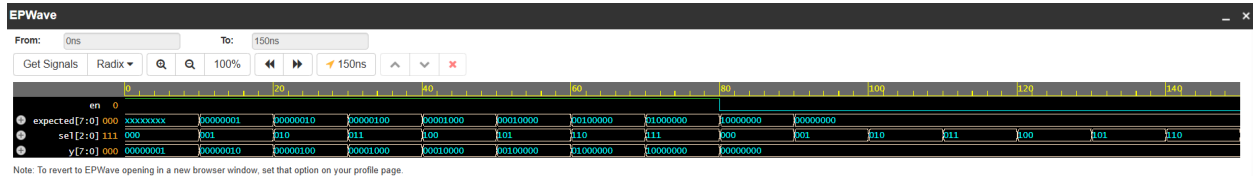
```

Task7_log_result2.png

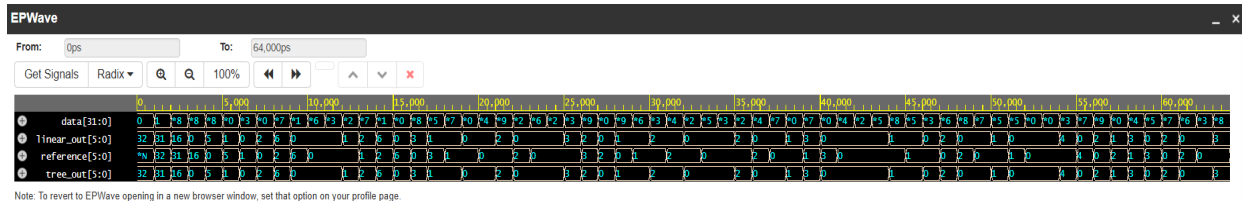
4-Waveform Screenshot:



Task6_decoder2to4_waveform.png



Task6_waveform.png



Task7_simple_waveform.png

5-Explanation:

Task7 Explanation:

The linear version has upto 32 logic levels because it checks each bit until he gets the 1 whereas the tree version has exactly 5 logic levels because it divides the 32 bit in such a way that base case/base level would be 2-bit group and from that we go to 4-bit, than 8-bit, than 16-bit and after that finally 32-bit level (which is 5th level from bottom to highest) after merging in each level.